

Project Design Phase

Solution Architecture

Date	19 September 2026
Team ID	
Project Name	ShopEZ — Apparel E-Commerce Application
Maximum Marks	4 Marks

Solution Architecture

Solution architecture is the process that bridges the gap between a business problem and its technical implementation. For ShopEZ, that means translating the problems identified in the Ideation Phase — shoppers needing accurate, variant-level stock visibility and a trustworthy checkout, and admins needing a unified way to manage products, orders, and store content — into a concrete, working technical structure.

Its goals for this project are to:

- Define the technology stack best suited to solving the identified problems (React + Vite, Express, MongoDB).
- Describe the structure, layers, and data flow of the application to any stakeholder reviewing the project.
- Define how each architectural layer maps to a specific feature and business requirement.
- Provide a reference that guided implementation from setup through to the completed application.

ShopEZ Solution Architecture Diagram

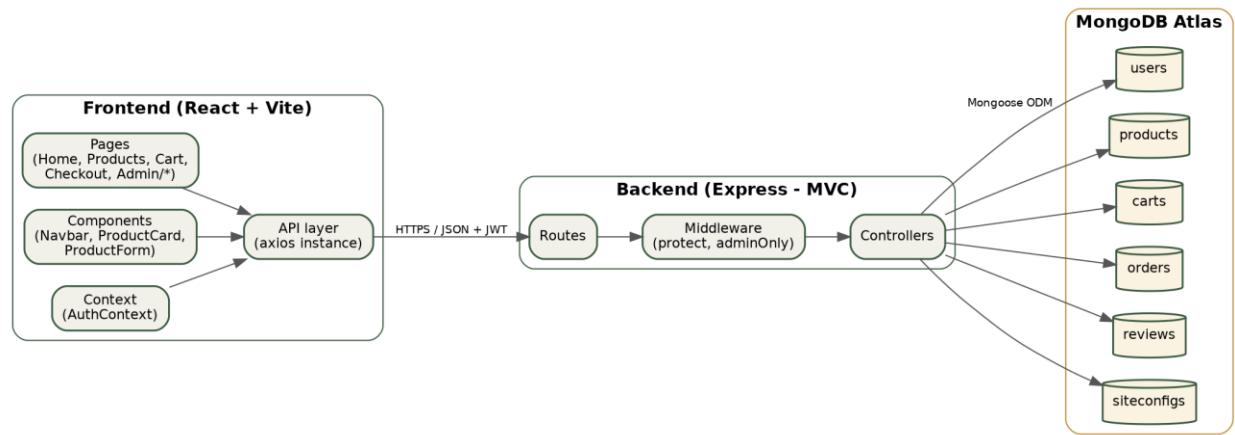


Figure 1: ShopEZ solution architecture — presentation, application, and data layers

How Each Layer Solves the Business Problem

- **Presentation Layer (React + Vite):** Delivers the category/search-based catalog browsing, per-variant size/color selection with live stock, cart, and checkout UI — directly addressing shoppers' need for clarity and speed.
- **Application Layer (Express — Routes, Middleware, Controllers):** Enforces business rules that a purely client-side app couldn't guarantee — re-validating stock at checkout time (not just at add-to-cart), hashing passwords, and gating every admin action behind JWT authentication and a role check.
- **Data Layer (MongoDB Atlas):** Stores stock at the individual variant level (not a single product-wide count) and snapshots order line items at purchase time, so admins always see accurate, real-time inventory and historical orders never drift when a product is later edited.
- **Admin Tooling (built on the same stack):** Because product, order, and site-configuration management live in the same codebase and data model as the storefront, admins get a single dashboard instead of a disconnected back-office tool — directly solving the fragmentation problem identified in the Ideation Phase.